KisanSetu: Product Overview and Workflow

KisanSetu is an app where farmers sell their crops directly to buyers, with no middleman. It works in Marathi, Hindi, and English, and it keeps working on a weak or broken internet connection.

Who uses it

- **Farmer:** lists crops and sets their own price.
- **Consumer:** a normal person buying vegetables or fruits for home.
- **Wholesaler, shopkeeper, restaurant:** buys in bulk and negotiates.
- **Admin and support:** checks listings, solves complaints, and keeps the platform safe.
- **Later:** delivery partners, collection centres, and quality inspectors.

The farmer's workflow

1. **Sign up:** the farmer enters a phone number and an OTP, picks a language, and adds their village location.
2. **Add a crop in simple steps:** take a photo, enter the crop name (they can speak it), then quantity, price, and pickup or delivery. The app shows the real market price for reference if verified data exists. If there is none, it says so and doesn't make one up.
3. **Works offline:** if there is no internet, the listing is saved on the phone as "pending". It uploads automatically when the network returns. The app never shows "published" until the server confirms.
4. **Get orders and offers:** the farmer sees new orders and buyer offers. They can accept, reject, or counter with a different price.
5. **Fulfil the order:** the farmer marks it accepted, packed, and then ready for pickup or shipped.
6. **Get paid and rated:** in the first version this is cash on delivery or pay on pickup. After the buyer confirms, the order is complete and the buyer can leave a review.

The buyer's workflow

1. **Browse and search:** buyers can search by crop (even by local names) and filter by price, quantity, quality, and distance. They can see nearby farmers and check seller profiles.
2. **Buy in one of two ways:**
    - **Normal purchase:** add to cart, choose delivery or pickup, and place the order.
    - **Negotiation:** send an offer or ask for a quote. The farmer can accept or counter, and when both agree, an order is created at that price.

3. **Chat:** buyer and farmer can message each other about a listing or an order. Messages are saved, so nothing is lost if the connection drops.
4. **Track and review:** the buyer follows the order status, confirms receipt, and leaves a review (only possible after a real completed order).

Protections built in

- **No overselling:** if only 10 kg is left and two people order at once, only one succeeds. The server checks stock, not the phone.
- **No double orders:** tapping "Place order" twice (or retrying on bad network) creates only one order.
- **Strict order steps:** an order moves in a fixed sequence (placed, accepted, confirmed, packed, shipped, delivered, completed). Nobody can skip steps.
- **Safety:** users can block and report others, and complaints go through a dispute process. Every admin action is logged.

Market price of the selected crop

The AI is not the source of the price. If you ask Gemini "what is today's onion price in Solapur?", it can give a confident but wrong number, and a farmer might price a whole harvest on it. So the price comes from real data, and the AI only explains it.

1. **A real data source supplies prices.** The usual option in India is government mandi (market yard) price data, the Agmarknet dataset published through data.gov.in. Its API access, update frequency, and licence terms have not been confirmed yet, so they will be checked before the app depends on it. The system is built so the source can be swapped, and the admin CSV import stays as a backup.
2. **A scheduled job fetches the prices** and saves them with the market name, unit, date observed, and source.
3. **When the farmer selects a crop**, the app looks up the nearest markets and shows, for example: "Solapur mandi: ₹1,800 to ₹2,400 per quintal, modal ₹2,100, as of 2 days ago, source: government data". If there is no recent data, it says so.
4. **Gemini only explains the numbers.** It reads the stored records and says things like "Prices are up 12% this week, and Pune pays more than Solapur." It never creates a price.

Revenue help for farmers

All of these use real stored prices and sales data. The farmer always decides.

- **Price suggestion:** a suggested range with the reasoning, such as "your quality grade, nearby mandi price, and current demand on the platform". The AI doesn't set the price.
- **Best place to sell:** compares nearby markets and shows where the same crop sells higher. It deducts a rough transport cost only if the farmer enters one.

- **Best time to sell:** shows historical price patterns for that crop. It says "past trend", never "guaranteed forecast".
- **Earnings preview:** gross price minus platform fee equals net earnings, shown before the farmer lists.
- **Bulk and tier pricing hints:** for example, "buyers often ask for 100 kg or more; consider a discount tier."
- **Unsold stock nudges:** if a listing gets views but no orders, it suggests a price check or a better photo.
- **Offer help:** when a buyer offers a price, it shows how that compares to the mandi price and suggests accept, counter, or reject, with the reason.

Revenue for the platform

- **Platform fee** on completed orders (configurable, 0% at the start).
- **Featured listings:** farmers pay to appear higher in search.
- **Premium subscriptions:** advanced price insights and analytics for farmers and wholesalers.
- **Business buyer plans:** procurement tools such as recurring orders and supplier comparison.
- **More sales through search:** the buyer search helper turns "100 kg onions near me" into filters, so buyers find stock faster and order more.

Paid features need a real payment gateway and some billing decisions, so they are added after the core marketplace works.

AI helpers

These are optional, and the app works fully without them. Without a Gemini key, farmers still see the raw price and trend numbers, just without the written explanation.

- **Market price explainer and revenue helper** (first priority): explains stored mandi prices, suggests price ranges, compares markets, and helps with offers, as described above. Every answer shows its source and date, and states that it is not a guaranteed forecast.
- **Buyer search helper:** a buyer types "I need 100 kg onions near me", and the AI turns that into search filters. The results still come from the real database.
- **Listing helper:** a farmer says "50 kg tomatoes, 20 rupees a kilo", and the AI fills in the form. It never invents prices or quantities, and it asks if something is missing.
- **Message translation:** it translates chats and clearly labels the text as machine-translated.
- The AI never places orders or changes anything by itself. The user always confirms.

Changes to the build plan

- Market price ingestion moves from post-MVP to **Phase 4**, so the price shown when listing is real.
- In Phase 8, price insights, sell-place and sell-time suggestions, and offer advice come first. The listing helper and translation come after.
- Paid features (featured listings, subscriptions, business plans) come after the core marketplace works.